Fiscal Year:	FY 2006	**Task Last Updated:**	FY 09/16/2005
PI Name:	Pishko, Michael V Ph.D.		
Project Title:	Microfabricated Optical Biosensor Arrays for in Situ Bioreactor Monitoring		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Cellular biotechnology & tissue engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	12/01/2003	**End Date:**	09/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	2
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	1		
COI Name (Institution):			
Grant/Contract No.:	NNJ04HB04G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	In this proposal a plan is forwarded whose primary focus is to develop multianalyte sensing technology to continuously and non-invasively monitor key cell culture media components in the bulk (using glucose, oxygen, lactate, pH, NO, and albumin as model analytes) within bioreactors. A collaborative working relationship has been established between the Department of Biomedical Engineering at Texas A&M University (TAMU) and the Chemical Engineering Department at Penn State University with combined expertise in optical sensing and imaging system design and biosensor and biomaterials synthesis. Specifically, we will immobilize analyte sensitive fluorophores and fluorophore-labeled biorecognition molecules (antibodies, lectins) within novel poly(ethylene glycol) based gel materials that are analyte permeable, non-fouling, maintain protein conformation and thus biological function. Furthermore, these gels can be micropatterned in a sensor array via photolithography. The gels will be designed to allow mass transfer of the analyte from the surrounding media into the gel where it can interact with an analyte-sensitive fluorophore or bind to the		

	recognition molecules.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	In this project period, we performed research related to pH sensing, oxygen sensing, and the sensing of reactive oxygen species in cell culture systems. In the area of pH sensing, we developed a pH sensor which utilizes SNARF-5F 5-(and -6)-carboxylic acid (SNARF-5F) entrapped in poly(ethylene) glycol (PEG) hydrogel microstructures prepared by photolithography. The intended use of this sensor is to detect real time cellular changes in pH in cell culture media without directly interacting with cells. The pH-sensitive fluorescence of the dye was characterized over a range of pH 5.5 to 9 in solutions of Phenol Red Dulbecco's modified eagle's medium (DMEM), Phenol Red free DMEM, and phosphate buffered saline (PBS). Hydrogel microstructures that encapsulated the dye were prepared were 500 um diameter and 1 mm in height cylinders on glass substrate. The poly(ethylene) glycol hydrogel were derived by crosslinking poly(ethylene) glycol diacrylate (molecular weight 575) with a photoinitiator using 365 nm ultraviolet light. The sensor response was linear for Phenol Red free DMEM and PBS in the pH range of 5.5 to 9 with no change in sensitivity between cell culture media and PBS. This research is described in a letter accepted for publication in Sensor Letters. In the area of oxygen sensing, we developed a dissolved oxygen sensor which utilizes dichlorotris(1,10-phenanthroline) ruthenium(II) hydrate nanoparticles entrapped in a poly(ethylene) glycol (PEG) hydrogel microstructure prepared by photolithography. The sensor can be used to detect cellular changes in dissolved oxygen in cell culture media without directly interacting with cells. The oxygen-sensitive phosphoresce of the dye was characterized in solutions of Phenol Red Dulbecco's modified eagle's medium (DMEM), Phenol Red free DMEM, and phosphate buffered saline (PBS). A dissolved oxygen range of 0.5 to 8 mg/L was detected in the solution and in the sensor. The phosphor nanoparticles were encapsulated in hydrogel microstructures that were 500 um diameter cylinders on glass substrate. The poly(ethylene) glycol hydrogel microstructures were derived by crosslinking poly(ethylene) glycol diacrylate (molecular weight 575) with a photoinitiator using 365 nm ultraviolet light. The sensor response was linear for Phenol Red free DMEM, Phenol Red DMEM and PBS and the range of the sensor was equal in these different solutions as well. Regarding sensors for reactive oxygen species, we studied the fabrication and characterization of poly(ethylene glycol) (PEG) hydrogel spheres containing the enzyme, horseradish peroxidase (HRP), for application as optical nanosensors for hydrogen peroxide. HRP was encapsulated in PEG hydrogel spheres by reverse emulsion photopolymerization, yielding spheres with a size range from 250 nm to 350 nm. Encapsulated HRP activity and sensitivity to hydrogen peroxide were investigated by the Amplex Red assay based on the fluorescence response as a function of H2O2. These HRP-loaded spheres were then introduced to murine macrophages with Amplex Red in the culture media. After phagocytosis, the biocompatibility of spheres was determined by live cell staining using calcein AM (5 microM). The HRP-loaded PEG hydrogel spheres were activated (i.e. fluorescent oxidized Amplex Red produced within the spheres) by oxidative stresses such as exogenous H2O2 (100 microM) and lipopolysaccharide (1ug/mL) which induced the production of endogenous peroxide inside macrophages. The results presented here indicate that after polymerization the enzyme activity of HRP was still maintained and that using these HRP-containing nanospheres, peroxide production could be sensed locally within cells. Results from this research were published in many peer-reviewed publications during the course of the award and three others have been submitted or accepted for publication. See Bibliography, as well as the below list: 1. Kim, S,-H.; Pishko, M. "Peptide-modified Fluorescent Silica Nanoparticles for Migrating Cell Analysis" submitted to Bioconjugate Chemistry. 2. Kim, S.-H.; Pishko, M. "Nanobiosensors for detecting intracellular reactive oxygen species at the single cell level" submitted to Anal. Chem.
Bibliography Type:	Description: (Last Updated: 08/24/2007)
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